

```

File Edit View Search Terminal Help
every 2.0s: free -m
Tue Feb 3 15:20:23 2015

              total      used      free     shared    buffers     cached
Mem:          3848        3004         844         304         164         1224
-/+ buffers/cache:      1615        2233
Swap:          3813           0        3813

```

Figure 1: The *watch* command

```

sn@bharatha:~/Public$ file Budget2014-15.pdf
Budget2014-15.pdf: MPEG ADTS, layer III, v2, 32 kbps, 22.05 klHz, Monoaural
sn@bharatha:~/Public$

```

Figure 2: The *file* command

This is particularly useful for admins in a multi-user system where the users may want to hide their MP3 or other audio/video files. It can even be used to test the file types in a Windows share by mounting the Windows share in Linux using *Samba*.

5. The *mtr* utility

The *mtr* utility (short for 'my traceroute') can be used as a network diagnostic tool, and combines the functionality of traceroute and ping utilities in one single program to provide a lot more statistical information. The tool checks the connection between the source (on which the command has been given) and the destination. A table of values is given as output and is refreshed continuously till the user presses *Ctrl-C*. The *-r* option gives a static report. Ideally, both ends of the network have to be tested using *mtr* to get a complete picture.

```
Smt 192.168.11.1
```

6. The *htop* tool

As per the man page, *htop* is an *ncurses*-based process viewer in Linux. It is an improvement over *top*. In fact, you may stop using *top* after getting used to *htop*. It allows you to scroll vertically and horizontally, so that you can see all the processes running on the system along with the commands that initiated the processes. Tasks can be filtered based on user accounts and a process can be killed within *htop*. This tool displays the output in colour, but the display format of the process details remains the same as in the output of the *top* command.

7. The *logsave* utility

The *logsave* utility captures the output of any command and stores it in a file along with the starting and ending time of the command. The unique aspect of this utility is that if the directory into which you are writing the output file does not exist, it is queued in memory and written after the directory is created. This is particularly useful for saving command outputs while booting up.

```
$logsave filelist.txt ls -l
```

The output of the *logsave* command is given below in Figure 5.

```

File Edit View Search Terminal Help
my traceroute [v0.85]
Tue Feb 3 15:16:35 2015
bharatha (0.0.0.0)
Resolver: Received error response 2. (server failure) of fields: quit

Packets
Hosts
 1. 192.168.1.224      0.0% 30 0.0 0.0 0.0 1.2 7.0
 2. 172.30.45.206     0.0% 30 13.0 52.3 5.7 185.3 50.5
 3. 172.30.45.194     0.0% 30 102.7 99.3 26.1 371.2 74.6
 4. 172.30.45.193     0.0% 37 174.6 133.1 43.8 330.7 83.7
 5. 192.168.11.1      0.0% 37 178.8 137.2 44.6 335.9 92.9

```

Figure 3: The *mtr* command

```

File Edit View Search Terminal Help
1 [ 2.0%
2 [ 0.0%
3 [ 0.7%
4 [ 15.7%
Mem [|||||] 1001/3040MB
Swap [|||||] 0/3813MB

Tasks: 120, 323 thr: 4 running
Load average: 0.30 0.22 0.22
Uptime: 00:51:04

3246 sn 20 0 1093M 530M 4324 5 1.3 24.2 4135.88 /usr/lib/firefox
7045 sn 20 0 30992 2136 1420 8 1.3 0.1 0:00.13 htop
1670 root 20 0 407M 101M 84376 5 1.3 2.6 443.56 /usr/bin/X -core
3235 sn 20 0 1093M 530M 4324 5 0.7 24.2 0:21.42 /usr/lib/firefox
2898 sn 20 0 1307M 204M 30008 0 0.6 3.3 3:23.30 /usr/bin/gnome-sh
1647 redis 20 0 10996 7324 944 5 0.0 0.2 0:04.31 /usr/bin/redis-se
6883 sn 20 0 924M 200M 19000 0 0.0 0.7 0:02.01 gnome-terminal
3236 sn 20 0 1393M 930M 4324 5 0.0 24.2 0:19.92 /usr/lib/firefox
3238 sn 20 0 1393M 930M 4324 5 0.0 24.2 0:19.99 /usr/lib/firefox
2786 sn 20 0 354M 6324 7888 5 0.0 0.2 0:11.70 /usr/bin/ibus-dae
2782 sn 20 0 354M 6324 7888 5 0.0 0.2 0:18.28 /usr/bin/ibus-dae
2822 sn 20 0 200M 3540 2804 5 0.0 0.1 0:05.42 /usr/lib/ibus/Ub
2910 sn 20 0 1997M 204M 30000 0 0.6 3.3 0:01.36 /usr/bin/gnome-sh

```

Figure 4: The *htop* command

8. The *look* command

The *look* command displays all the lines in a file that start with a particular string. The input file should be sorted as *look* performs a binary search. If a file name is not mentioned, then by default, the */usr/share/dict/words* file is used. Only alphanumeric characters are compared and the *-f* option ignores the case of the letters of the alphabet.

```
$look "printf" avltree.c
```

The *grep* command can be used to do the same task, but the performance of *look* is better. Its other use is that it can check the spelling of a word. Type the first few letters of any word that you want the spelling of and the command will display all the words starting with those letters.

```

sn@bharatha:~$ cat listing.txt
log of ls -l
Tue Feb 3 15:27:02 2015

total 68
-rw-rw-r-- 1 sn sn 910 Dec 10 11:19 Comments.txt
drwxr-xr-x 2 sn sn 4096 Feb 3 12:35 cprgs
drwxr-xr-x 3 sn sn 4096 Feb 2 15:47 Desktop
drwxr-xr-x 14 sn sn 4096 Feb 3 14:41 Documents
drwxr-xr-x 4 sn sn 4096 Jan 29 16:41 Downloads
drwxr-xr-x 2 sn sn 4096 Dec 8 17:05 dwheeler
-rw-rw-r-- 1 sn sn 8980 Oct 15 21:00 examples.desktop
-rw-rw-r-- 1 sn sn 40 Feb 3 15:27 listing.txt
drwxr-xr-x 2 sn sn 4096 Jan 23 16:26 Music
-rw-rw-r-- 1 sn sn 832 Feb 3 12:02 output.log
drwxr-xr-x 4 sn sn 4096 Feb 3 15:57 Pictures
drwxr-xr-x 4 sn sn 4096 Feb 3 15:52 Public
drwxr-xr-x 2 sn sn 4096 Oct 15 21:05 Templates
drwxr-xr-x 2 sn sn 4096 Jan 19 12:00 trial
drwxr-xr-x 4 sn sn 4096 Jan 31 12:19 Videos

Tue Feb 3 15:27:02 2015
-----
sn@bharatha:~$

```

Figure 5: The *look* command